EPI 203: Reproducible Report

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Introduction

In this analysis, we

Note: there are many things that we are taking creative liberty on, including things like the units of health care costs and what the overall purpose of this study is.

Methods

```
df |>
    tbl_summary(
        statistic = list(all_continuous() ~ "{mean} ({sd})")
    )
```

Study Population.

Our study sample contains 5000 individuals, 13% of which are female, 13% are smokers. The mean age is 45. The average cost of health care is $9398 per year.

Outcome.

Exposures.

Statistical Analysis.

We ran a generalized linear regression to assess the relationship of `age` and `cost`, adjusting for `female`, `smoke`, and `cardiac`. `age` was categorized into three groups: 18-29, 30-49, and 50+ for better visualization and interpretability.

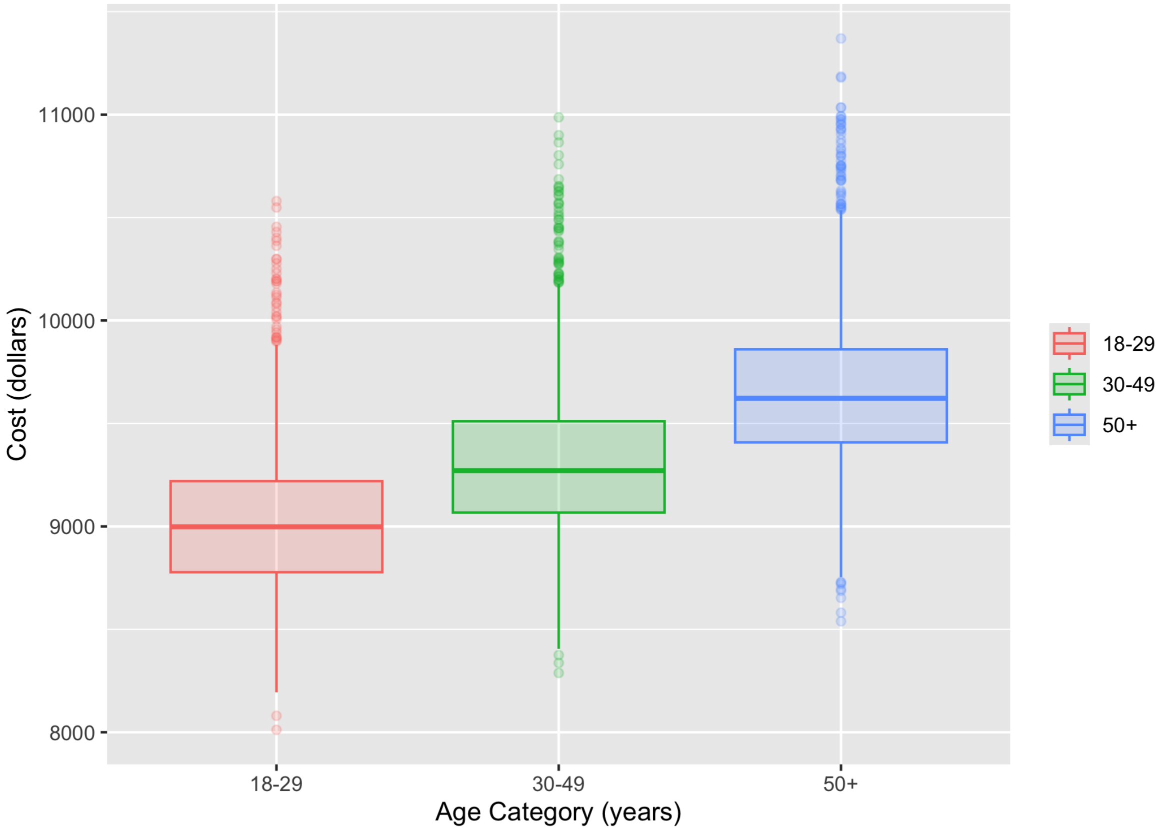
Characteristic	**N = 5,000**[1]
smoke	646 (13%)
female	2,866 (57%)
age	45 (16)
cardiac	
cardiac event	275 (5.5%)
no cardiac event	4,725 (95%)
cost	9,398 (448)
age_cat	
18-29	1,078 (22%)
30-49	1,841 (37%)
50+	2,081 (42%)

[1]n (%); Mean (SD)

Results

From our linear model, we see that age is a significant and positive predictor of `cost` and that individuals in older age groups are more likely to pay more `cost` than younger age groups (on average, there seems to be ~200-300 increase in `cost` per age group). Figure 1 shows the distribution of `cost` by `age_cat`.

```
fig1 <-
    df |>
    ggplot() +
    geom_boxplot(aes(x = age_cat, y = cost, color = age_cat, fill = age_cat),
                 alpha = 0.2) +
    labs(x = "Age Category (years)",
         y = "Cost (dollars)",
         color = NULL, fill = NULL)
```



Discussion

AI Statement: *I attest that I did not use generative AI technology to complete this assignment.*